LESSON 5.6
PYTHAGOREAN THEOREM

LESSON INTRODUCTION

MA.8.GR.1.1 Apply the Pythagorean theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.

LEARNING TARGETS

- I can apply the Pythagorean theorem to find the hypotenuse of a triangle.
- I can apply the Pythagorean theorem to find the leg of a triangle.

BENCHMARK COHERENCE

BUILDING ON

MA.6.GR.1.2 Find distances between ordered pairs, limited to the same *x*-coordinate or the same *y*-coordinate, represented on the coordinate plane.

WORKING TOWARD

MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean theorem.

ESSENTIAL QUESTIONS

- How can I apply what I know about Pythagorean theorem to find the missing sides of a triangle?

LESSON NARRATIVE

In this lesson, students apply what they learned about the converse of the Pythagorean theorem in previous lessons to build an understanding of the Pythagorean theorem. Students begin by understanding which side of a triangle is the hypotenuse. Students then find the length of a hypotenuse and then move on to finding length of a missing leg of a right triangle using the Pythagorean theorem.

COMPONENT SUMMARY

5.6.1 WARM-UP *(~5 MIN.)*

Compare and contrast equations

5.6.2 REFRESHER *(5 MIN.)*

Review of right triangles

5.6.3 GUIDED INSTRUCTION *(10 MIN.)*

Use Pythagorean theorem to solve for hypotenuse

5.6.4 TRY IT *(5-10 MIN.)*

Practice finding the hypotenuse

5.6.5 COLLABORATION *(5 MIN.)*

Comparing right triangles

5.6.6 GUIDED INSTRUCTION *(10 MIN.)*

Solve for the missing leg using the Pythagorean theorem

5.6.7 YOUR TURN! *(5-10 MIN.)*

Practice solving for the missing leg

5.6.8 WRAP-UP *(~5 MIN.)*

Self-reflection on this lesson

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RESOURCES

GLOSSARY

Key Terms:

- Hypotenuse
- Converse of the Pythagorean theorem

Additional Terms:

- Pythagorean theorem
- Irrational number

MATERIALS

- (Optional) Patty paper
- (Optional) Highlighters
- Calculators

ADDITIONAL PREPARATION

- This is a vocabulary-rich lesson. Consider preparing your classroom-specific vocabulary or glossary strategies for support (word wall, math cards, etc.).

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may forget to square each number.
- Students may substitute the numbers in the incorrect spot when using the Pythagorean theorem.
- Students may square numbers incorrectly. Make sure they are squaring the sides and not multiplying by two.
- Students may forget to take the square root at the last step of the Pythagorean theorem.

THINGS TO CONSIDER

- Consider having students color-code the sides of the triangle a, b and c, each corresponding to a certain color.
- Determine if your students need the support of all or part of the 5.6.2 Refresher.
- If time is an issue, then consider assigning 5.6.7 Your Turn! for homework.
- In problems that require several steps, students should not round until the final step. Rounding midstep could significantly change the answer. If not given a place value for rounding, then determine what place value you would like your students to round to. Many assessments have students round to three decimal values.

LITERACY AND LANGUAGE ROUTINES

Which One Doesn't Belong?

The 5.6.1 Warm-Up engages students in a Which One Doesn't Belong? routine. Encourage students to note the similarities between the equations presented and the theorems learned in the previous lesson. Help students draw connections and consider the mathematical reasons each equation could "not belong."

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

In this lesson, students are provided an opportunity to relate previously learned concepts to new concepts. Students should relate what they know about the converse of the Pythagorean theorem to the Pythagorean theorem. Students use the relevant details to create a plan to solve for a missing side. Students can use the similarities in Lessons 4 and 5 to connect concepts in this lesson.

DIFFERENTIATE INSTRUCTION

This lesson hinges upon student understanding of right triangles and Pythagorean theorem. While there are plenty of visuals provided as students work through each section, providing kinesthetic entry points may benefit students who struggle to label the features of triangles presented differently (for example, flipped or reversed). Consider providing triangle manipulatives for students to use to help determine the hypotenuse and right angles as they use the formula to determine the missing side lengths of the triangles throughout this lesson.

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5.6.1 WARM-UP: WHICH ONE DOESN'T BELONG?

TEACHER MOVES

Which One Doesn't Belong?

Notice that each expression could "not belong" for a different reason. The purpose of this activity is for students to define terms carefully and use words precisely in order to explain their thinking. Encourage students to explain their thinking with a partner and foster student discussion to encourage mathematical reasoning as it relates to the theorems discussed in the previous lesson.

5.6.1 Warm-Up: Which One Doesn't Belong?

1. Which one doesn't belong?
   A. $3^2 + b^2 = 5^2$
   B. $b^2 = 5^2 - 3^2$
   C. $3^2 + 5^2 = b^2$
   D. $3^2 + 4^2 = 5^2$

   Answers will vary.

2. Explain your thinking.
   Answers will vary. Answer choice A does not belong because it is the only answer choice where the variable is on the same side of the equal sign as another term. Answer choice B does not belong because it is the only answer choice with subtraction. Answer choice C does not belong because the value of b is not a whole number. Answer choice D does not belong because it is the only answer choice without variables.

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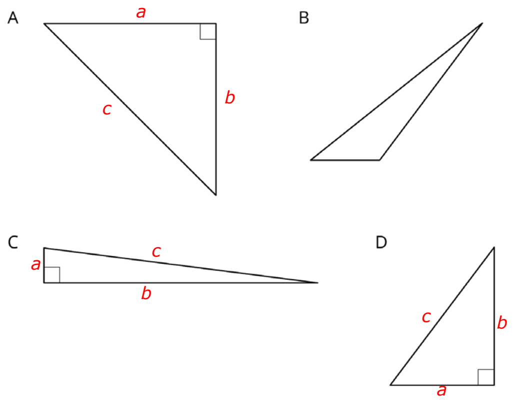
5.6.2 REFRESHER: PARTS OF A RIGHT TRIANGLE AND USING SQUARE ROOTS

5.6.2 Refresher: Parts of a Right Triangle and Using Square Roots

The ***hypotenuse*** is the longest side of a right triangle; the side opposite the right angle.

The hypotenuse can be found directly across the right angle.

1. For each right triangle, label the legs a and b and the hypotenuse with c.



The ***Converse of the Pythagorean Theorem*** states that if the lengths a, b, and c of the three sides of a triangle satisfy the relationship $a^2 + b^2 = c^2$, then the triangle is a right triangle.

DIFFERENTIATE INSTRUCTION

Consider having students highlight or color-code the different sides of the right triangle. They can do this throughout the lesson to help visualize the different sides. For example, the hypotenuse is always highlighted yellow.

2. Solve for the positive value of the missing variable in each equation. If necessary, approximate your answer.

   a. $c^2 = 64$
   8

   b. $d^2 = 100 + 21$
   11

3. What strategy did you use to find each missing value?
   Answers will vary. I found the square root of the given values.

TEACHER MOVES

Students may need a reminder on using square roots to solve this equation. Consider spending a few minutes reviewing that, as students will need to apply this idea throughout the lesson. Remind students that there are both positive and negative values that will solve each of these, but the question only wants the positive value. Explain that this is because triangles cannot have negative side lengths.

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5.6.3 GUIDED INSTRUCTION: SOLVING FOR THE HYPOTENUSE USING THE PYTHAGOREAN THEOREM

DIFFERENTIATE INSTRUCTION

When the orientation of the triangle is flipped, students may have a hard time identifying the hypotenuse. Consider having students trace the triangle onto patty paper and rotating the triangle in different ways. Consider also highlighting the right-angle symbol in the triangle.

TEACHER MOVES

If students answer question 1b with both theorems, probe for their reasoning. A student could state that they know possible side lengths of the hypotenuse from the two given side lengths. This would not be an incorrect response.

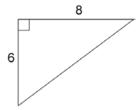
ENRICHMENT

Encourage students to check their work to ensure it is reasonable to build the habit of considering reasonableness throughout this lesson and unit.

- How can we check if our answer is reasonable? (Triangle inequality theorem.)
- How can you apply the triangle inequality theorem to determine if the answer is reasonable?

5.6.3 Guided Instruction: Solving for the Hypotenuse Using the Pythagorean Theorem

1. A right triangle with side lengths 6 and 8 is shown.



a. Which side length of the triangle is unknown?

○ hypotenuse
○ legs

b. Which theorem applies specifically to the side lengths of right triangles?

○ Pythagorean Theorem
○ Triangle Inequality Theorem

c. Use the Pythagorean Theorem to substitute the given side lengths for the appropriate variables.

$$a^2 + b^2 = c^2$$
$$6^2 + 8^2 = c^2$$

d. Solve the equation for the value of the missing variable.
10

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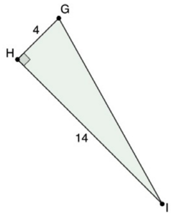
2. Right triangle GHI is shown with side lengths 4 and 14 is shown.



Find the length of the hypotenuse using the Pythagorean Theorem. If the measure of the side length is an irrational value, write an exact solution and an approximate solution.

a. State Pythagorean Theorem $a^2 + b^2 = c^2$

b. Substitute $4^2 + 14^2 = c^2$

c. Solve for the missing variable. $16 + 196 = c^2$

$212 = c^2$

d. Solutions

Exact solution:

$c = \sqrt{212}$

Approximate answer:

$c \approx 14.56$

5.6.3 GUIDED INSTRUCTION: SOLVING FOR THE HYPOTENUSE USING THE PYTHAGOREAN THEOREM (CONTINUED)

TEACHER MOVES

A common misconception is that it matters which side length is labeled "a" and which is labeled "b." Use the commutative property to show students that it does not matter which leg is labeled. Emphasize that it would matter if the wrong length was used for "c." Since students often substitute the numbers in the incorrect spot in the equation for Pythagorean theorem, emphasizing the importance of correctly labeling the hypotenuse could help students alleviate this error.

If you notice students mixing up the side lengths as they substitute them in the formula, provide guidance through questioning.

- Which side represents the hypotenuse? Which side is across from the right angle?
- Does it matter which side length is labeled a or b in this instance? Why or why not?
- Which side lengths can you substitute for a and b in the formula?

Consider having students continue to highlight the sides in different colors. They can then highlight the corresponding variable in the Pythagorean theorem that matches the side. This will help students be sure they are substituting the numbers in the correct spots.

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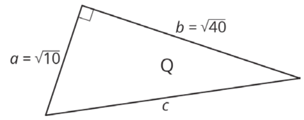
5.6.3 GUIDED INSTRUCTION: SOLVING FOR THE HYPOTENUSE USING THE PYTHAGOREAN THEOREM (CONTINUED)

TEACHER MOVES

Students worked with square roots in Unit 2. The question spirals what a student learned in that unit by emphasizing that the length is an irrational number. Use this question to refresh students' memory on what an irrational number is as well as how to square a square root. Students should not use an approximate value for $\sqrt{10}$ or $\sqrt{40}$. Also, students should not have to rewrite $\sqrt{50}$ as $5\sqrt{2}$. The original purpose of rewriting radicals was to be able to use a square root table to determine an approximate value. Square root tables are no longer necessary because students have access to calculators.

3. A right triangle is shown with side lengths $\sqrt{10}$ and $\sqrt{40}$.



Find the length of the hypotenuse using the Pythagorean Theorem. If the measure of the side length is an irrational value, write an exact solution and an approximate solution.

a. State Pythagorean Theorem

$$a^2 + b^2 = c^2$$

b. Substitute

$$(\sqrt{10})^2 + (\sqrt{40})^2 = c^2$$

c. Solve for the missing variable.

$$10 + 40 = c^2$$

$$50 = c^2$$

d. Solutions

Exact solution:

$$c = \sqrt{50}$$

Approximate answer:

$$c \approx 7.071$$

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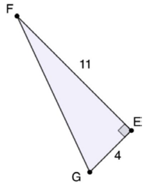
5.6.4 Try It: Finding the Hypotenuse

1. Right triangle FEG is shown with side lengths 4 and 11.



Find the length of the hypotenuse using the Pythagorean Theorem. If the measure of the side length is an irrational value, write an exact solution and an approximate solution.

a. Which side length is unknown?	○ hypotenuse ○ leg
b. State Pythagorean Theorem.	$a^2 + b^2 = c^2$
c. Which variable in the equation is unknown?	○ a ○ b ○ c
d. Substitute values into equation.	$4^2 + 11^2 = c^2$
e. Algebraically solve for missing variable.	$16 + 121 = c^2$ $137 = c^2$
f. Solutions	$c = \sqrt{137}$ $c \approx 11.705$

5.6.4 TRY IT: FINDING THE HYPOTENUSE

DIFFERENTIATE INSTRUCTION

Consider posting the Pythagorean theorem on the board for students to reference.

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5.6.5 COLLABORATION: WHAT'S THE LEG?

DIFFERENTIATE INSTRUCTION

If students are having a hard time comparing the two triangles, then consider the following:

- Consider having students trace the triangles on patty paper and rotate so that they have the same orientation and they are easier to compare.
- Students can also highlight the corresponding side lengths for a, b, and c to help them visually compare the triangles.

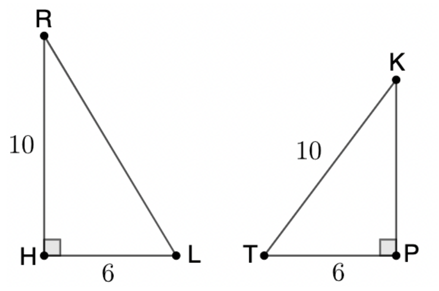
TEACHER MOVES

Think-Pair-Share

Have students think about questions 1a and 1b independently first before pairing up and talking with a partner. After students have come up with some similarities and differences, have them share out with the class. Discussing these questions as a class will help students synthesize the material as they create equations using the Pythagorean theorem for the first time.

5.6.5 Collaboration: What's the Leg?

1. Triangles RHL and TKP are shown. Both triangles have one side with an unknown measurement.



   a. How are these triangles similar?
   Answers will vary. These triangles are both right triangles. The shortest leg of both of these triangles measures 6 units.

   b. How are these triangles different?
   Answers will vary. Triangle RHL is bigger than Triangle KPT. The missing side on Triangle RHL is the hypotenuse and the missing side on Triangle KPT is the leg.

   c. Work with your partner to write equations to determine the length of the missing side in each right triangle using the Pythagorean Theorem.

Triangle RHL	Triangle TKP
$6^2 + 10^2 = c^2$	$6^2 + b^2 = 10^2$

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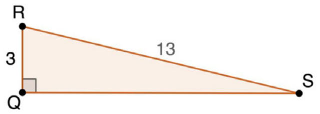
5.6.6 Guided Instruction: Solving for a Missing Leg Using the Pythagorean Theorem

The Pythagorean Theorem can also be used to find a missing leg in a right triangle.

1. Use the Pythagorean Theorem to find the approximate measure of $\overline{QS}$.



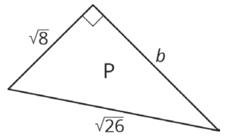
a. State the Pythagorean Theorem.
$a^2 + b^2 = c^2$

b. Substitute the length of the [○ hypotenuse / ○ legs] for c.

c. Substitute the given leg length for [○ a or b. / ○ a or c. / ○ b or c.]

d. Solve the equation for the missing variable.
$QS \approx 12.649$ or $QS = \sqrt{160}$

2. Find the exact measure of b in the diagram.



a. State the Pythagorean Theorem.
$a^2 + b^2 = c^2$

5.6.6 GUIDED INSTRUCTION: SOLVING FOR A MISSING LEG USING THE PYTHAGOREAN THEOREM

TEACHER MOVES

For both questions 1 and 2, notice if students are incorrectly substituting the values into the Pythagorean theorem. If students are struggling to substitute values correctly, provide guidance through questioning.

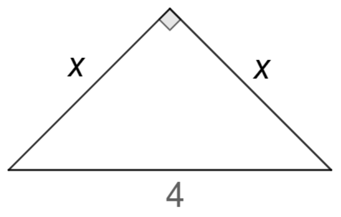
- Which side represents the hypotenuse? Which side is across from the right angle?
- Does it matter which side length is labeled a or b in this instance? Why or why not?
- Which side length can you substitute for a and b in the formula?

b. Substitute the length of the [○ hypotenuse / ○ leg] for c.

c. Substitute the given leg length for [○ a or b. / ○ a or c. / ○ b or c.]

d. Solve the equation for the missing variable.
$b = \sqrt{18}$

3. Find the exact length of x in the diagram using the Pythagorean Theorem.



a. State the Pythagorean Theorem.
$a^2 + b^2 = c^2$

b. Substitute the length of the [○ hypotenuse / ○ leg] for c.

c. Substitute the given leg length for [○ a and b. / ○ a and c. / ○ b and c.]

d. Solve the equation for the missing variable.
$x = \sqrt{8}$

TEACHER MOVES

Consider a conversation about how the information given on this triangle is different from the last two.

- What does it mean if both sides are labeled x?
- How does that change the way we will solve?
- Will the values of x be greater or less than 4? How do you know?
- Which side represents the hypotenuse? How do you know?

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5.6.7 YOUR TURN!

DIFFERENTIATE INSTRUCTION

Consider creating task cards for students to match using this activity. Students need to match the triangle with the correct equation.

QUESTIONING

For students who finish early, challenge them to also solve the equations for the missing side.

ENRICHMENT

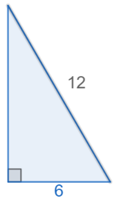
Students may struggle getting started with question 2, as it is complicated by the division of the larger triangle into two right triangles. Scaffold through questioning.

- Which side do we need to solve for first? Why?
- What information do you already know?
- What information can you gather from what you already know?
- What side length represents the hypotenuse of the entire triangle? How is this helpful in finding the length of y?

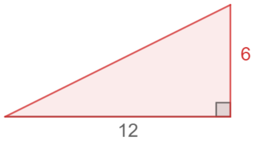
5.6.7 Your Turn!

1. Match the triangle to the appropriate equation that could be used to solve for the missing side.

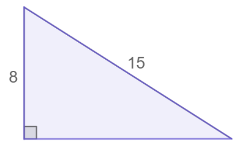
A. *III*



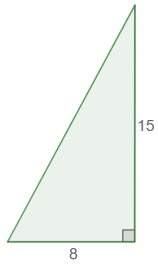
B. *II*



C. *I*



D. *IV*



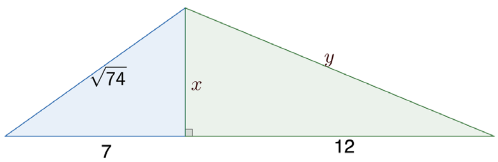
I. $8^2 + b^2 = 15^2$

II. $12^2 + 6^2 = c^2$

III. $a^2 + 6^2 = 12^2$

IV. $8^2 + 15^2 = c^2$

2. Find the measures of x and y in the diagram using the Pythagorean Theorem.



$x = 5$; $y = 13$

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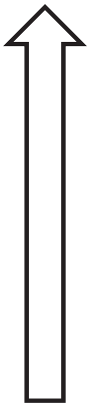
5.6.8 Wrap-Up: Reflection

1. Color in the arrow up to the statement which best describes your current understanding.
Answers will vary.



I'm so confident, I can explain this to someone else!

I can get to the right answer, but I don't understand well enough to explain it yet.

I understand some of this, but I don't understand all of it yet.

I tried hard and listened, but I am finding this challenging. I will make sure I get help with this next lesson.

I do not understand any of this yet. There are things I could do to be a better learner next lesson.

5.6.8 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

To gain a better understanding of students' comprehension, consider having students go back and either highlighting or circling the part of the lesson that was the most challenging and the part of the lesson that was the easiest for them.

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